Ukrainian Catholic University

AI Course · Semester Project Proposal · 2026

World Model for Autonomous Driving: Which Scene Representation Is the Best?

Team:

Yurii Bobas

Vladyslav Vasylchenko

Oleh Hombosh

Mentor:

Marta Sumyk

Project proposal submitted for the "Artificial Intelligence" course

in the

Faculty of Applied Sciences

2026-09-20

1. Problem and motivation

Autonomous vehicles must anticipate how a traffic scene will evolve over the next few seconds in order to act safely. **World models** address this by predicting future scene states from past observations and a planned vehicle maneuver (**ego action**). This requires encoding the scene in some representation, and the literature offers several competing approaches — **object-centric** (discrete agents with position, class, motion), **BEV/occupancy** (explicit bird’s-eye-view grid), and **latent BEV** (the same grid compressed by an autoencoder) — each pursued in isolation, with no controlled comparison between them under a shared protocol. This matters because the choice of representation is a foundational design decision for any world model, yet it is currently made on intuition or convenience rather than evidence: practitioners have no empirical basis for choosing between a compact, interpretable representation and a dense or learned one. The question interests us specifically because it sits at the intersection of representation learning and generative modeling — it is not just an engineering choice but a question about what kind of structure a model needs to reason about physical space and causality, and nuScenes’ well-known quirks (skew toward trivial scenarios, the ego-status shortcut) make it a genuinely tricky benchmark to get right rather than a routine training exercise.

2. Related work

According to the survey by author [1], world models for autonomous driving are generative spatial-temporal models. They compress multi-sensor inputs into a compact latent state and predict how it changes over time. The survey groups these methods by scene representation and points out that pure BEV gives a lightweight 2D view, but it loses vertical geometry in complex scenes. This leads to methods like object-centric models or latent BEV. For example, BEVWorld [2] encodes sensor inputs into a compact latent BEV space and uses a diffusion model for prediction. Because it generates all future frames at once instead of one by one, it avoids accumulating autoregressive errors. At the same time, Drive-WM [3] shows why it is important to evaluate predictions using planning metrics (like L2 error and collision rate) and stress tests, such as shifting the ego-vehicle position out of the training distribution. Since previous works do not directly compare these setups under the same test conditions, this project focuses on building a reproducible benchmark to compare object-centric, BEV, and latent BEV representations for accuracy and stability, as well as an interactive demo.

3. Data

We use nuScenes [4], a public large-scale dataset for autonomous driving developed by Motional. It enables researchers to study challenging urban driving situations using the full sensor suite of a real self-driving car. It contains 1,000 scenes of 20 s each, collected in two cities: Boston and Singapore, and includes data from 6 cameras, 1 LiDAR, and 5 radar sensors, together with detailed maps and vehicle pose information. The scenes are manually selected to show a diverse and interesting set of driving maneuvers, traffic situations, and unexpected behaviors, with left- and right-hand traffic. The full dataset includes approximately 1,400,000 camera images, 390,000 LiDAR sweeps, 1,400,000 radar sweeps, and 1,400,000 object bounding boxes. The data are split into 700 training, 150 validation, and 150 test scenes; annotations are provided for the training and validation sets, while the test set does not come with object annotations and is intended for evaluation through the official benchmark. The dataset is available free of charge for non-commercial use under the CC BY-NC-SA 4.0 license, subject to additional terms specified by Motional.

4. Approach and methods

To compare the three scene representations for next-frame prediction, we build a unified evaluation pipeline on the nuScenes dataset. All three encoders share the same LiDAR and multi-camera fusion backbone based on the BEVWorld tokenizer, and they differ only in how they represent the scene internally. Each version predicts $N = 6$ future frames (a 3-second horizon) from $P = 3$ past frames in a single pass, which avoids

autoregressive error build-up. We test all three approaches using the same metrics: FID and Chamfer distance to measure reconstruction and prediction quality at 1s and 3s horizons, L2 deviation and collision rate at 1s, 2s, and 3s for planning stability, and inference latency along with peak memory per run to check if they can work in real-time RTOS environments. We also run the out-of-distribution (OOD) stress test from Drive-WM [3], shifting the car sideways by 0.5 meters to see how much the accuracy drops in unusual situations. The main deliverables of this work are a reproducible benchmark script that runs all three representations under this setup and an interactive demo that shows predicted future scenes side by side with the ground truth.

5. Evaluation

We will evaluate the three scene representations using the evaluation principles used in autonomous-driving world models described in a survey [1]. Prediction accuracy will be measured primarily by Chamfer Distance (CD), which measures the similarity between a predicted point cloud and its ground-truth counterpart; lower CD values indicate more accurate predictions. To evaluate prediction stability, we will report the prediction error at multiple future horizons (e.g., 1, 2, and 3 seconds) and measure how rapidly the error increases with the prediction horizon. This is particularly relevant for world models, where errors may accumulate during multi-step future prediction [2]. As a simple baseline, we will use a persistence model that predicts the next state to be identical to the current state. All three representations will be evaluated on the same scenes, prediction horizons, and train/validation/test split to ensure a fair comparison. We expect the three representations to differ in prediction accuracy and stability, especially as the prediction horizon increases.

6. Plan and risks

Before midterm: define the evaluation protocol — precise definitions of the three scene representations (object-centric, BEV/occupancy, latent BEV), a shared evaluation space, the metric set, and the success criterion for comparing representations; set up the nuScenes data pipeline and convert all three representations into the shared evaluation format; implement mandatory trivial baselines (constant-velocity extrapolation, last-frame repetition); and implement the object-centric and BEV/occupancy predictors on a shared backbone architecture, producing a first round of results against the baselines.

Before the final: implement the latent BEV predictor on the same shared backbone; run the full evaluation of all three representations across forecasting horizons; conduct the action-sensitivity test, comparing forecasts under different ego actions from identical states; build the interactive demo visualizing counterfactual forecasts; and finalize documentation for reproducibility (configs, seeds) alongside the final report.

Risks: misleading conclusions from dataset skew without trivial baselines; insufficient compute, especially for latent BEV diffusion, biasing results toward undertraining rather than representation quality; confounded comparisons if pretrained third-party checkpoints are used instead of a shared backbone; incomplete/outdated reference code, extending reproduction time; undefined success criterion, risking metrics without a defensible conclusion.

7. Contribution statement

- **Yurii Bobas** — evaluation protocol design, object-centric representation & predictor, action-sensitivity testing
- **Vladyslav Vasylchenko** — nuScenes data pipeline, shared evaluation format & baselines, BEV / occupancy predictor
- **Oleh Hombosh** — latent BEV predictor, full evaluation & metrics analysis, interactive demo & documentation

References

[1] Tuo Feng, Wenguan Wang, and Yi Yang. A survey of world models for autonomous driving. *arXiv preprint arXiv:2501.11260*, 2025.

[2] Yumeng Zhang, Shi Gong, Kaixin Xiong, Xiaoqing Ye, Xiaofan Li, Xiao Tan, Fan Wang, Jizhou Huang, Hua Wu, and Haifeng Wang. Bevworld: A multimodal world simulator for autonomous driving via scene-level bev latents. *arXiv preprint arXiv:2407.05679*, 2024.

[3] Yuqi Wang, Jiawei He, Lue Fan, Hongxin Li, Yuntao Chen, and Zhaoxiang Zhang. Driving into the future: Multiview visual forecasting and planning with world model for autonomous driving. *arXiv preprint arXiv:2311.17918*, 2023.

[4] Holger Caesar, Varun Bankiti, Alex H. Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. nuscenes: A multimodal dataset for autonomous driving. *arXiv preprint arXiv:1903.11027*, 2019.